

Literature status: the $j = 2$ Santaló / many-body Blaschke–Santaló case

Run `fa_banach_001`; source arXiv:2203.14815

Source question

Kalantzopoulos and Saroglou’s paper *On a j -Santaló Conjecture* formulates, for $2 \leq j \leq k$, a sharp product-volume conjecture for tuples of symmetric convex bodies satisfying an $\mathcal{E}_j$ -polarity condition. In the case $j = 2$, this condition is

$$\mathcal{S}_2(x_1, \dots, x_k) = \sum_{1 \leq i < \ell \leq k} \langle x_i, x_\ell \rangle \leq \binom{k}{2} \quad (x_i \in K_i),$$

and the conjectured conclusion is

$$\prod_{i=1}^k |K_i| \leq |B_2^n|^k.$$

The source also records the functional version and notes that for $j = 2$ it is the Kolesnikov–Werner many-function Blaschke–Santaló conjecture.

Supporting literature

Chen, Li, Xi, and Xu’s later paper *The many-body Blaschke–Santaló type inequality via optimal transport*, arXiv:2606.30579, explicitly identifies the $j = 2$ geometric problem as the many-body Blaschke–Santaló inequality proposed by Kalantzopoulos and Saroglou. Its abstract says that the authors prove the sharp inequality and, using the geometric–functional equivalence of Kalantzopoulos and Saroglou, establish the Kolesnikov–Werner functional version.

The literature answer

Theorem 1 (Chen–Li–Xi–Xu, Theorem 1.1). *Let $n \geq 1$, $k \geq 2$, and let $K_1, \dots, K_k \subset \mathbb{R}^n$ be origin-symmetric measurable sets of finite volume. If*

$$\mathcal{S}_2(x_1, \dots, x_k) \leq \binom{k}{2} \quad \text{for all } x_i \in K_i,$$

then

$$\prod_{i=1}^k |K_i| \leq |B^n|^k.$$

The paper also characterizes the equality cases.

This proves the $j = 2$ case of the geometric j -Santaló conjecture from arXiv:2203.14815, since $B_j^n = B_2^n$ for $j = 2$ and $\mathcal{E}_2 = \mathcal{S}_2/\binom{k}{2}$. It is slightly stronger than the convex-body formulation because the supporting theorem allows measurable sets of finite volume.

By the geometric–functional equivalence proved in arXiv:2203.14815, Proposition 4.1, the same supporting paper obtains the functional many-body Blaschke–Santaló inequality. This is recorded as Theorem 1.2 of arXiv:2606.30579 and answers the Kolesnikov–Werner $j = 2$ functional conjecture.

Scope and limitations

This is a literature-status packet, not an original proof from the run. The supporting authors explicitly know the relation: they name the inequality as the one proposed by Kalantzopoulos–Saroglou and cite their geometric–functional equivalence.

The scope is the $j = 2$ pairwise-interaction case only. The higher $j > 2$ conjectures in arXiv:2203.14815 remain outside this packet, except for the classes already proved in the source paper itself, such as $j = k$ and the unconditional cases.

Verification notes

Before packaging, the run lightweight indexes were searched for arXiv:2203.14815, Santaló/Santalo keywords, the title phrase, Kolesnikov–Werner, many-function Blaschke–Santaló, and related convex-geometric terms. No existing exact packet for this target was found.

Web search on 2026-07-03 found arXiv:2606.30579, posted June 29, 2026. The arXiv page and supporting PDF state Theorems 1.1 and 1.2 and explicitly identify the source problem and functional consequence.

Human review recommendation

Record the $j = 2$ component of arXiv:2203.14815 as already answered by arXiv:2606.30579. Keep the higher j -Santaló conjectures separate.

References

- [1] P. Kalantzopoulos and C. Saroglou, *On a j -Santaló Conjecture*, arXiv:2203.14815.
- [2] S. Chen, Y. Li, D. Xi, and Z.-F. Xu, *The many-body Blaschke–Santaló type inequality via optimal transport*, arXiv:2606.30579.
- [3] A. V. Kolesnikov and E. M. Werner, *Blaschke–Santaló inequality for many functions and geodesic barycenters of measures*, arXiv:2010.00135.